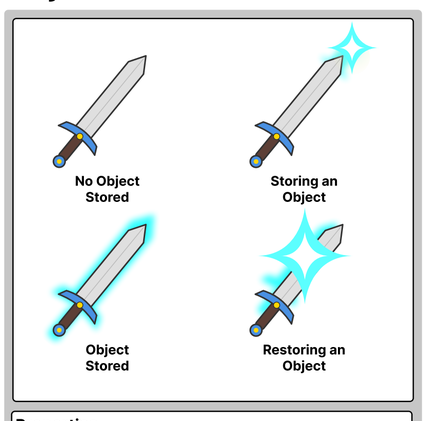
Top-Down 2D Dungeon Crawler
By Owen Haedrich

Object: Time Sword



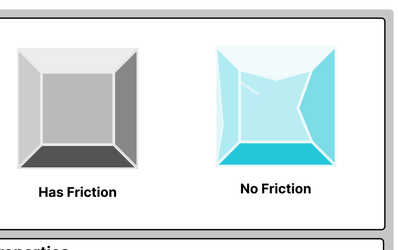
Properties
- Stored Object: The last object hit by the *Time Sword* and the position of that object when it was hit.

Behaviours
- Store an Object: Hit an object to store its position.
  - Fails if the object is immovable (e.g. a wall).
- Restore the Object: Return the stored object to its stored position, and clear the stored object from the *Time Sword*.
  - Fails to activate if the stored position in occupied.

Game State Feedback
- Visual
  - Store an object: Animation and VFX
    - Flash from stored object to sword.
  - An object is stored: Shape and VFX
    - Aura around sword while an object is stored.
  - Restore an object: Animation and VFX
    - Flash from sword towards stored object.
  - Fail to activate: VFX
    - Sword flashes red.
- Audio
  - Store an Object
  - Restore an Object
  - Fail to activate

Systemic Feedback
- Moves other objects in the game environment.

Object: Push-Block



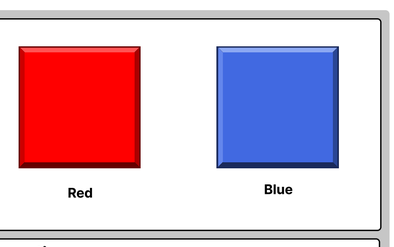
Properties
- Friction: Either has friction or has no friction.

Behaviours
- Slide: Try to move along either the X- or Y-axis when pushed by the player or hit by another moving object.
  - Has Friction: Move while the player is pushing it.
  - No Friction: Slide until it collides with something.
    - Will slide over and past pressure plates.

Game State Feedback
- Visual
  - Friction: Colour and Shape
    - The two permutations have different visuals.
  - Slide: VFX
    - Dust behind when sliding.
    - Dust in front when stopping.
- Audio
  - Start sliding
  - While sliding
  - Stop sliding

Systemic Feedback
- Moves in the game environment.

Object: Pressure Plate



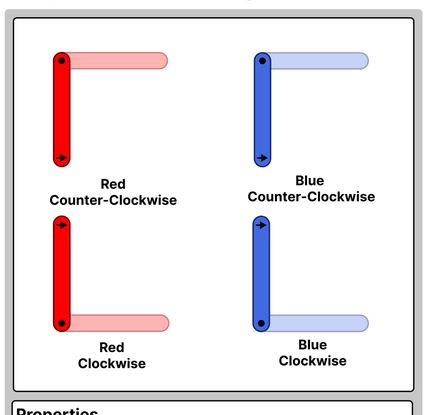
Properties
- Toggle State: Either active or inactive.
- Colour: Either Red or Blue.

Behaviours
- Toggle: Change toggle state when the player character or a *Push-Block* is on top of the *Pressure Plate*.

Game State Feedback
- Visual
  - Toggle: VFX
    - Coloured flash when activated.
  - Toggle State: Shape
    - Coloured aura around the plate when active.
- Audio
  - Activated
  - Deactivated

Systemic Feedback
- Toggles objects in the game environment.

Object: Rotating Barrier



Properties
- Toggle State: Either active or inactive.
  - Open when active, closed when inactive.
- Colour: Either Red or Blue.
- Direction: Either clockwise or counter-clockwise.

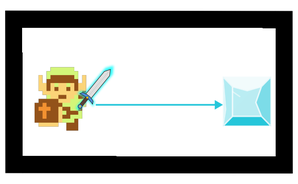
Behaviours
- Rotate: Quickly swing towards the target position (open or closed). Stop if colliding with an immovable object. Pushes the player and other moveable objects.

Game State Feedback
- Visual
  - Toggle State: Shape
    - Rotation direction arrow and inactive position highlight changes with the state.
- Audio
  - Start rotating
  - While rotating
  - Stop rotating

Systemic Feedback
- Moves in the game environment.
- Moves other objects in the game environment.

Simple Example: Time Sword and Push-Block with no Friction

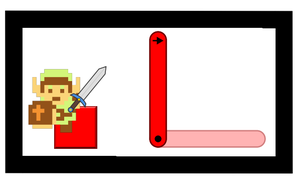
Store the Push-Block



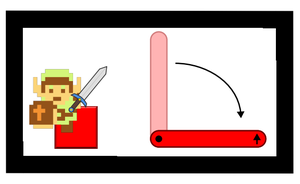
Slide the Push-Block

Restore the Push-Block

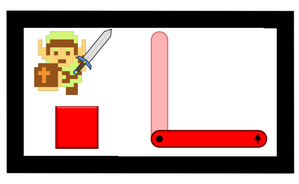
Simple Example: Pressure Plate and Rotating Barrier



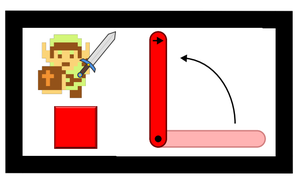
Step on Pressure Plate



Same-Colour Rotating Barriers Move to Open Position

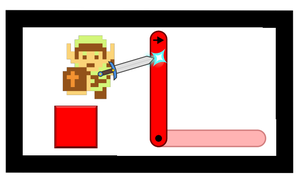


Step off Pressure Plate

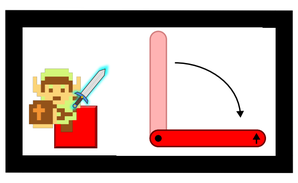


Same-Colour Rotating Barriers Move to Closed Position

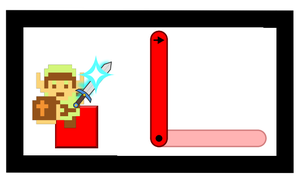
Simple Example: Time Sword with Pressure Plate and Rotating Barrier



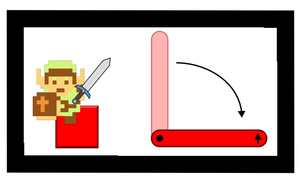
Store the Rotating Barrier



Step on the Pressure Plate and the Barrier Rotates to its Open Position

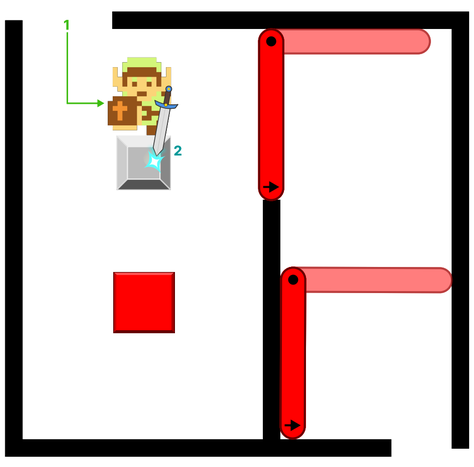


Restore the Rotating Barrier

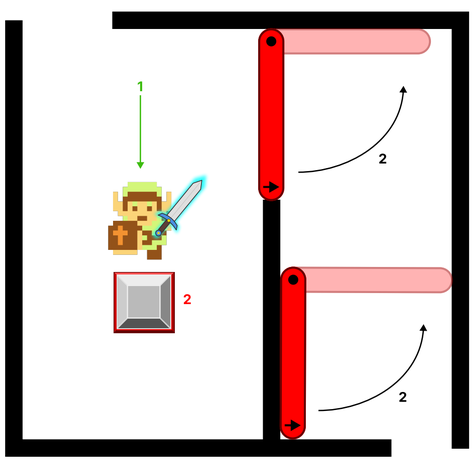


Rotating Barrier is Still Active so it Rotates back to its Open Position

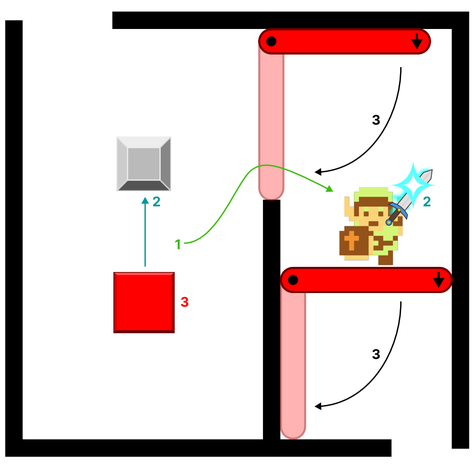
Example Scenario #1



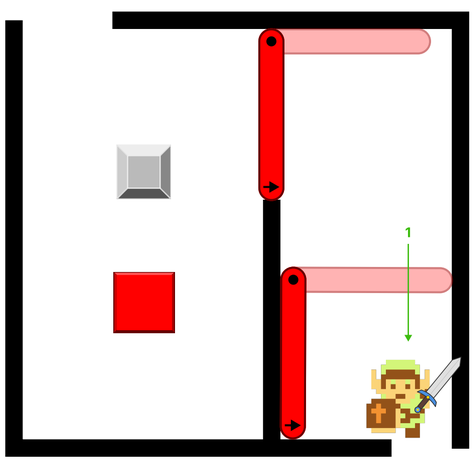
A
1. Enter the room and approach the *Push-Block*.
2. Use the *Time Sword* to store the *Push-Block* while it is off the *Pressure Plate*.



B
1. Push the *Push-Block* to activate the *Pressure Plate*.
2. The *Pressure Plate* will activate both *Rotating Barriers* and they will move to their open position.

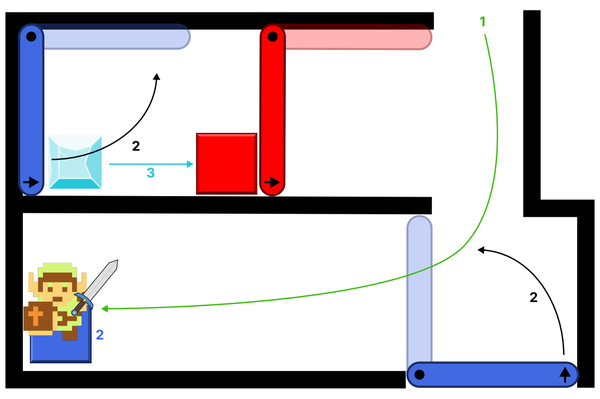


C
1. Move between the two *Rotating Barriers*.
2. Activate the *Time Sword* to move the *Push-Block* off of the *Pressure Plate*.
3. The *Pressure Plate* is now inactive and the *Rotating Barriers* will return to their closed positions.

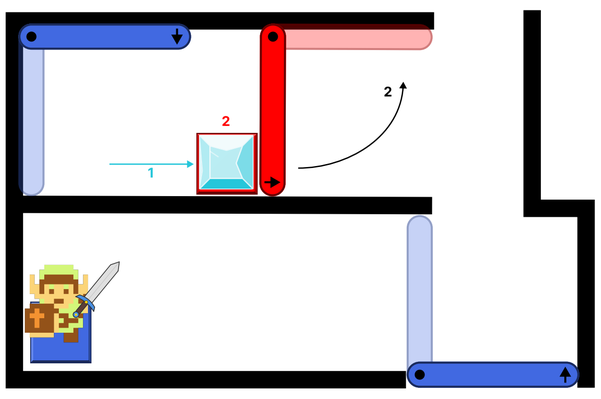


D
1. With the path clear, walk to the exit. The room is complete.

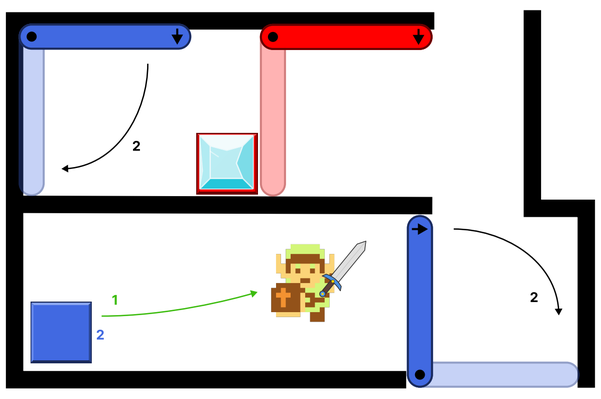
Example Scenario #2



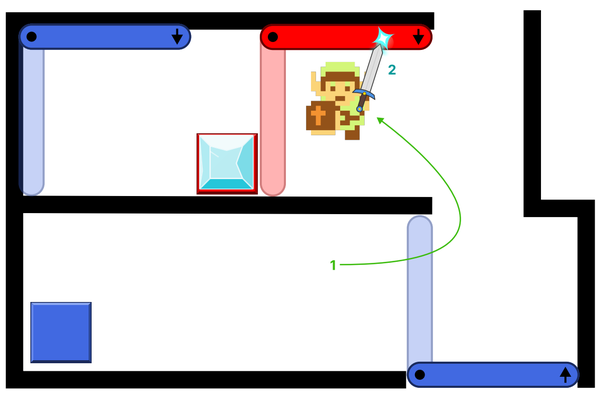
A
1. Enter the room and approach the blue *Pressure Plate*.
2. Activate the blue *Pressure Plate* to open the blue *Rotating Barriers*.
3. The *Push-Block* with no friction is pushed by the *Rotating Barrier*.



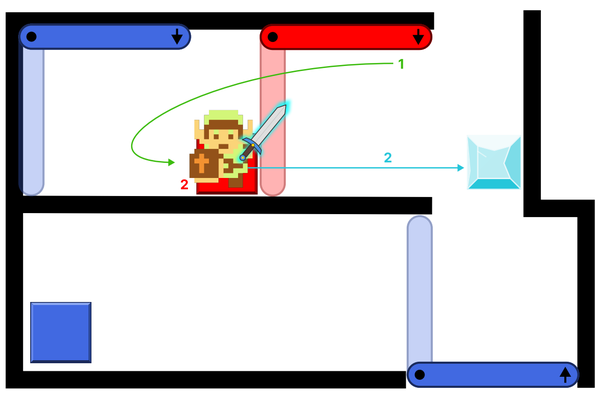
B
1. The *Push-Block* with no friction slides until it hits the red *Rotating Barrier*.
2. The *Push-Block* activates the red *Pressure Plate*, opening the red *Rotating Barrier*.



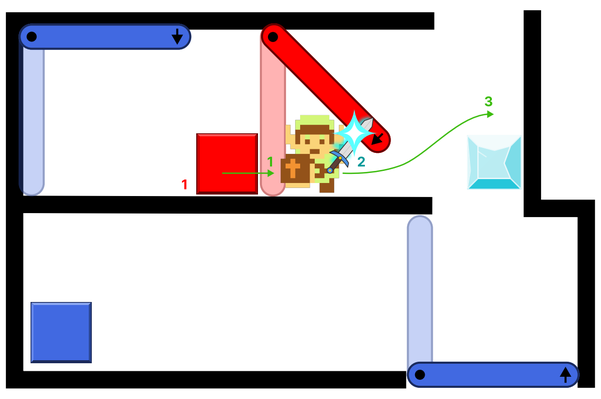
C
1. Step off the blue *Pressure Plate*.
2. With the blue *Pressure Plate* clear, the blue *Rotating Barriers* return to their closed positions.



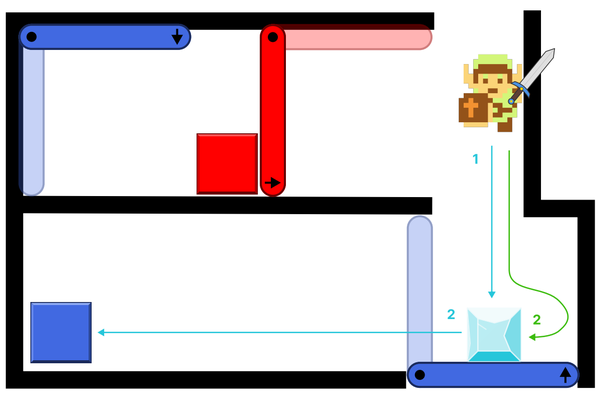
D
1. Approach the red *Rotating Barrier*.
2. Store the red *Rotating Barrier* in its open position with the *Time Sword*.



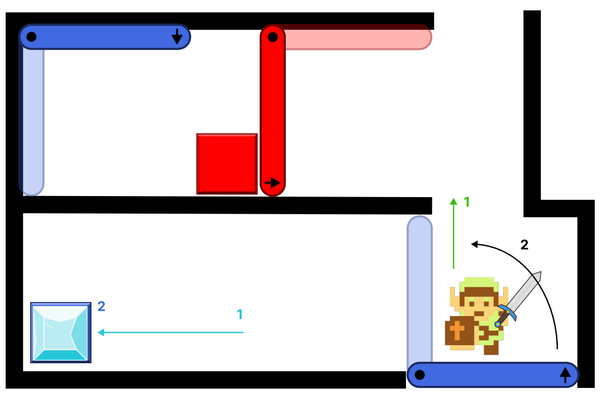
E
1. Approach the *Push-Block* and push it from the left side.
2. Step on the red *Pressure Plate* to prevent the the *Rotating Barrier* from closing on the *Push-Block* while it slides.



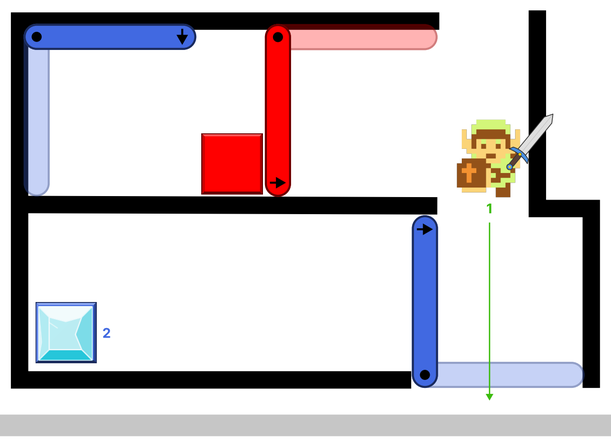
F
1. Step off the red *Pressure Plate* and walk toward the *Push-Block*. The red *Rotating Barrier* will start to close.
2. Just before being hit by the *Rotating Barrier*, activate the Time Sword to reset the *Rotating Barrier* to its open position.
3. Walk to the top side of the *Push-Block*.



G
1. Push the *Push-Block* downwards.
2. Approach the *Push-Block* and push it to the left.



H
1. While the *Push-Block* is sliding, walk out of the range of the blue *Rotating Barrier*.
2. The *Push-Block* activates the *Pressure Plate* and the blue *Rotating Barrier* opens.



I
1. With the path clear, walk to the exit. The room is complete.

Assignment 2:
Systemic Games with Emergent Systems

Course: GAME 10017 Game Systems Design

Professor: Raphaël Tétreault

Designer: Owen Haedrich

Date: 2026/02/19

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Q1. Justify Objects, Properties, Behaviours, and Relationships

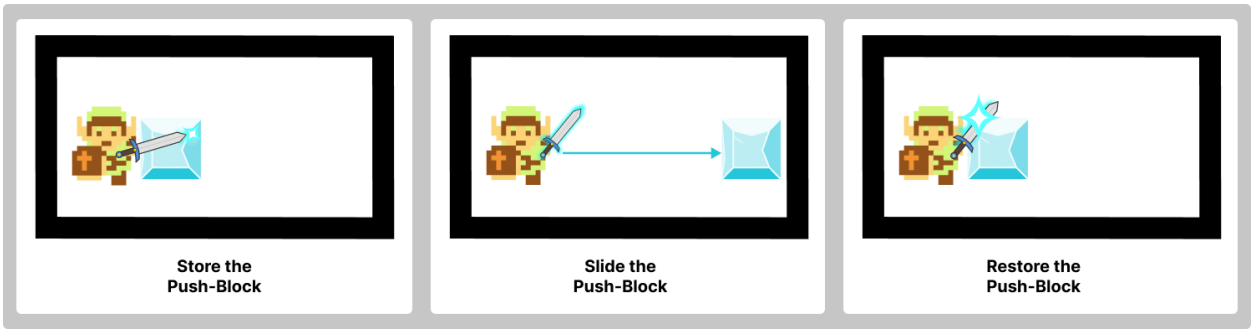
For each mechanic, justify why you designed these specific objects, properties, behaviours, and relationships.

1.0 Assumptions

For this design, I made some assumptions about the game, the player character, and common object properties:

- There are walls, a moving player, and other basic top-down dungeon-crawler elements
- The player can reset their current room at any time (i.e. if they get stuck, they can return to the state where they first entered the room)
- Objects have common properties such as position and rotation that are unlisted in the one-page design document

1.1 Time Sword



Simple Example: Time Sword and Push-Block with no Friction

This object is designed to create and trigger coupled and context-dependent interactions between the player and the game environment.

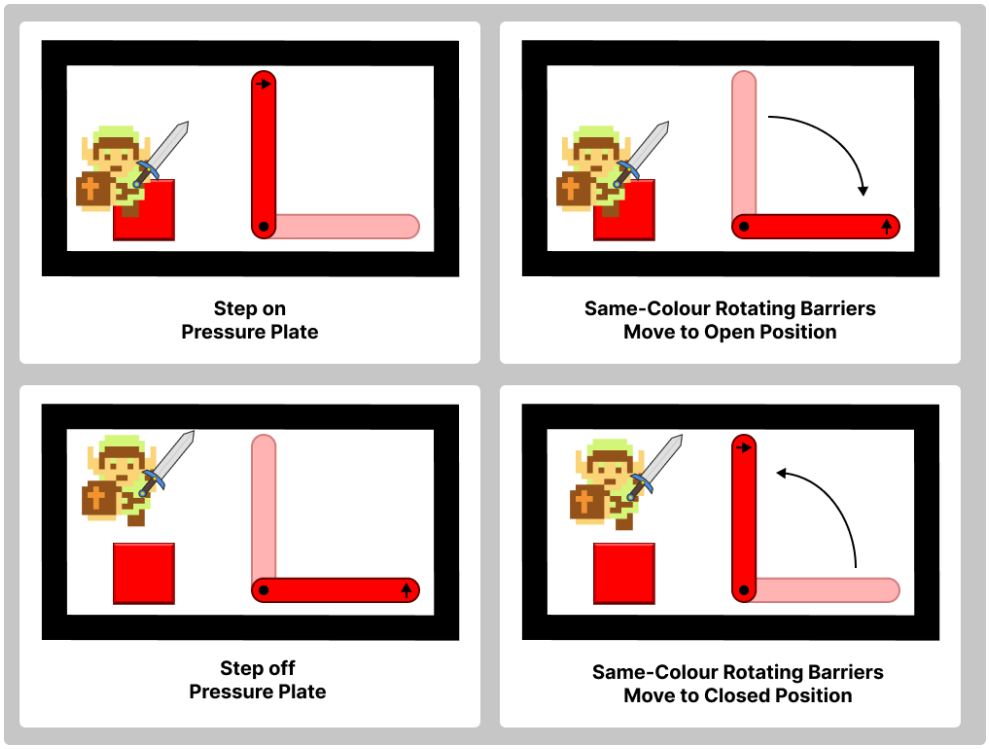
The main property of the *Time Sword* is a slot for storing a snapshot of a moveable object in a particular position. The snapshot can be restored, instantly returning the stored object to the stored position.

The *Time Sword* has a direct relationship with the *Push-Block* and *Rotating Barrier*, where it can store and restore their positions. It also has an indirect relationship with the *Pressure Plate*, where restoring the position of a *Push-Block* can activate or deactivate a plate.

Its two key behaviours, storing and restoring the position of other objects, create a wide possibility space for game states and player decisions. The player can manipulate objects to save one in the desired position, then, they can move and manipulate the level until they find the right time to restore that object.

Since the *Time Sword* is held by the player, they can restore the stored object at any time from any position in the game environment. However, they can only reposition that object once before storing it again, and they can only store one object at a time. Accordingly, the *Time Sword* is a key source of player agency in this design.

1.2 Pressure Plate



Simple Example: Pressure Plate and Rotating Barrier

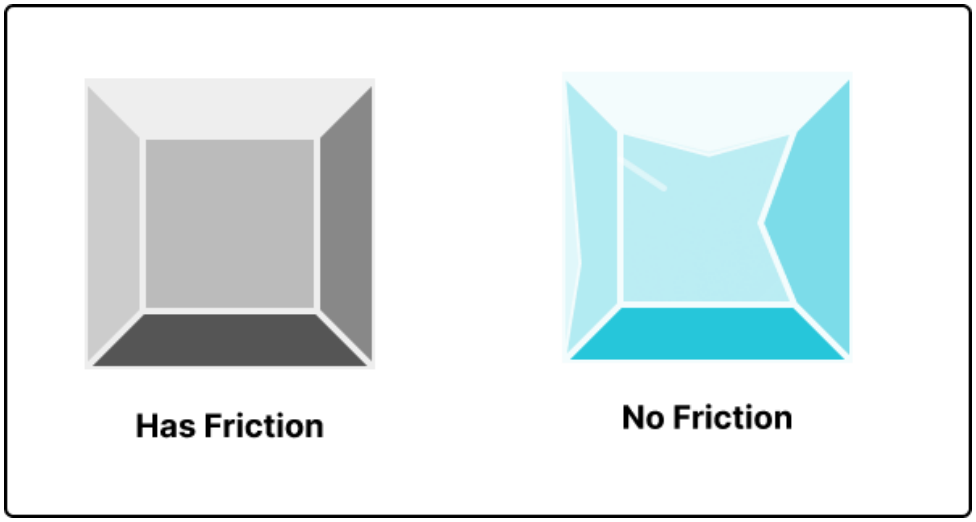
The *Pressure Plate* is the central object to puzzle solving in this design. It creates coupled interactions between itself, the player, and the game environment.

The main property of the *Pressure Plate* that the player interacts with is its toggled state. It is active when the player character or a *Push-Block* are on top of the plate, and inactive otherwise. Its other property, its colour, connects the *Pressure Plate* to *Rotating Barriers* that follow its toggled state. When a *Pressure Plate* is activated, *Rotating Barriers* of the same colour are also activated. This behaviour allows the player to manipulate the game environment and make their way to the exit doors.

This design has exactly two *Pressure Plate* colours to facilitate Rube Goldberg-style chain reactions, without the complexity becoming cumbersome. The first additional colour adds significant depth, but each additional colour would have diminishing returns.

The *Pressure Plate* has direct relationships with the *Push-Block* and the *Rotating Barrier*. And as discussed above, it creates context-dependent interactions with the *Time Sword*. Depending on the level configuration and positioning of the active game objects, triggering a *Pressure Plate* can create a cascading sequence of interactions, including the activation of more *Pressure Plates* or the deactivation of itself. This makes *Pressure Plates* a building block for interesting gameplay scenarios and essential to several systems in this design.

1.3 Push-Block



Push-Block Permutations

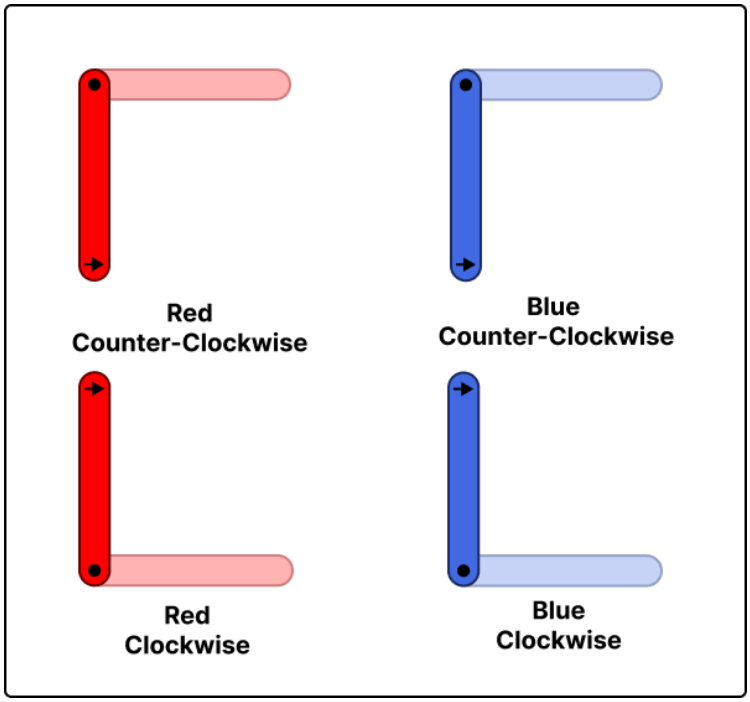
The *Push-Block* is an object that can be moved by the player. These blocks can be pushed either vertically or horizontally.

Push-Blocks have direct relationships with all the other objects defined in this design. Their position can be stored and restored by the *Time Sword*, they can activate *Pressure Plates*, and they can be pushed by *Rotating Barriers*. They are also the object that the player can manipulate the most, since they can be moved freely across the floor, transported by the *Time Sword*, and pushed by *Rotating Barriers*.

An interesting property of this object is friction. There are two permutations of the *Push-Block* based on this property: with friction and frictionless. These permutations have different behaviour when pushed. *Push-Blocks* with friction will only travel while being pushed, while the frictionless variant will travel until it has a collision (e.g. it hits a wall or another *Push-Block*). These two variants of behaviour increase the possible number of emergent gameplay scenarios, without significantly increasing the mental overhead for players. To illustrate, friction allows the player to easily position a block anywhere in a room, but a block without friction can travel a long distance while the player positions themselves or another block.

The *Push-Block* becomes even more interesting when considering how the two permutations can interact with each other, for example a *Push-Block* with friction could be positioned in-line with a *Push-Block* with no friction to stop the frictionless block from sliding all the way across a room.

1.4 Rotating Barrier



Rotating Barrier Permutations

This object acts like a door that rotates between two positions 90 degrees apart. The barrier can be stopped while rotating if it collides with an immovable object, such as a wall or a *Push-Block* that can't be moved to a new valid position.

The *Rotating Barrier* has four basic permutations determined by its predetermined properties: colour (red and blue), and rotation direction (clockwise and counterclockwise). These properties, combined with the in-level position and rotation of the barriers, facilitate many possible level configurations. Additionally, the 90 degree rotation behaviour of the *Rotating Barrier* can open one path while closing another, creating dynamic paths in the level layout.

To continue, *Rotating Barriers* have direct relationships with all the other objects: they can be stored with the *Time Sword*, they can be activated by *Pressure Plates*, and they can push *Push-Blocks*. Additionally, it has an interesting relationship with the player character: it will push them while rotating. It's also important to note that the barriers close too fast for the player to walk through if they are stepping off the *Pressure Plate* that is holding it open. Therefore, they might have to time their movement with a sliding *Push-Block* or their *Time Sword* activation. This introduces real-time challenges with on top of the logical puzzle solving aspects of this design.

Q2. Justify Emergence, Gameplay Possibility Space

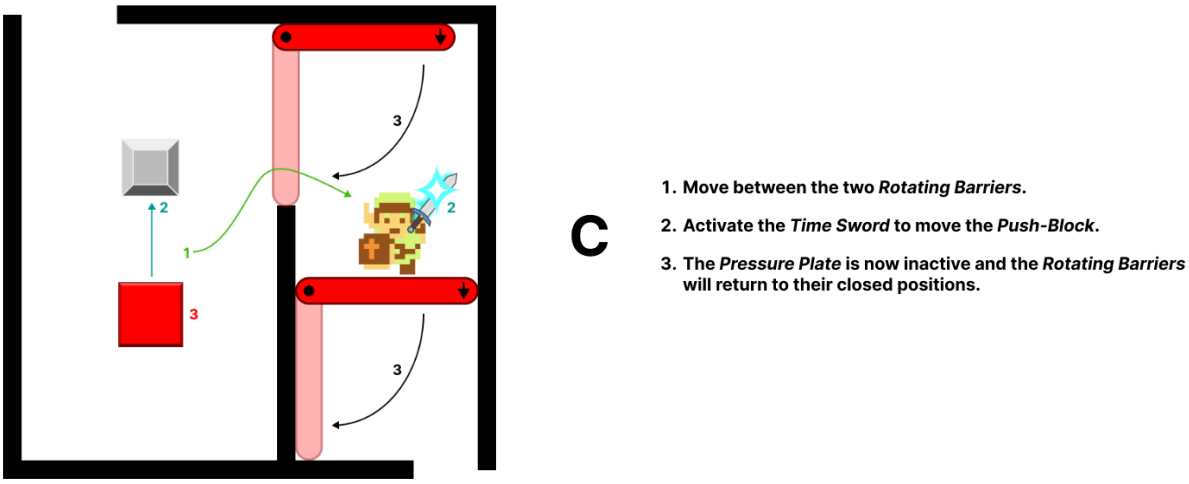
Overall, justify how you used a minimal number of objects with a sufficient number of properties to generate a wide possibility space.

This design is focused on creating a wide possibility space through emergent interactions between a small collection of objects with minimal properties.

The objects defined in this design are heavily interconnected. As discussed above, the *Push-Block* and the *Rotating Barrier* are directly related to every other object and can interact with other instances of themselves.

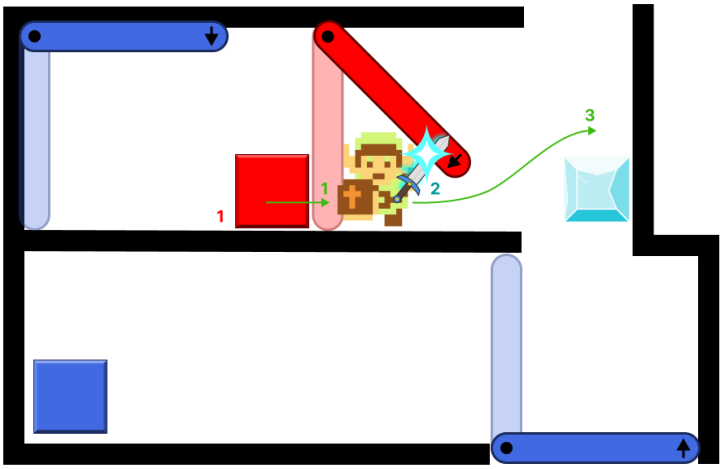
The *Time Sword* and the *Pressure Plate* are not directly related to one another, but they can influence each other through second-order interactions via their relationships with the *Push-Blocks* and *Rotating Barriers*.

Through these overlapping relationships, complexity emerges. To illustrate, in the first scenario, we see a simple cascading logical sequence in part C. Activating the *Time Sword* moves the *Push-Block* off the *Pressure Plate*, causing it to become inactive. The *Pressure Plate* toggles the target position of the *Rotating Barrier* which opens the path to the exit.



Scenario #1: Part C

Furthermore, in the second scenario we can see how these elements can create real-time challenges for the player. There are two instances of these real-time, context-dependent interactions in the scenario. First, in part F, the player needs to use the *Time Sword* on the *Rotating Barrier* to create enough time for themselves to walk through.

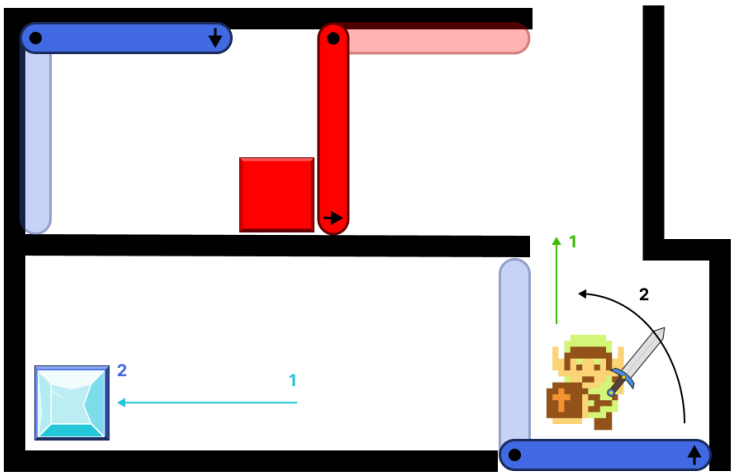


F

1. **Step off the red *Pressure Plate* and walk toward the *Push-Block*. The red *Rotating Barrier* will start to close.**
2. **Just before being hit by the *Rotating Barrier*, activate the Time Sword to reset the *Rotating Barrier* to its open position.**
3. **Walk to the top side of the *Push-Block*.**

Scenario #2: Part F

Here, the player needs to assess their walk speed with respect to the speed of the closing *Rotating Barrier* to determine the correct time to activate the *Time Sword* so they can pass through. Second, in part H, the player needs to move while the *Push-Block* is sliding so they don't get trapped.

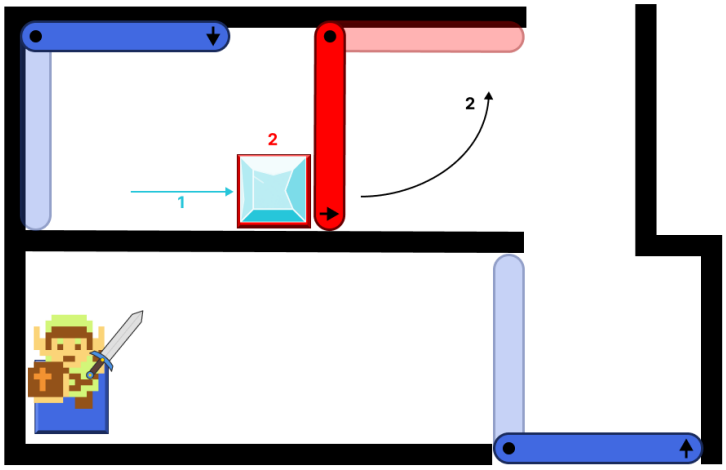


H

1. **While the *Push-Block* is sliding, walk out of the range of the blue *Rotating Barrier*.**
2. **The *Push-Block* activates the *Pressure Plate* and the blue *Rotating Barrier* opens.**

Scenario #2: Part H

Beyond the interactions explored in the scenarios, we can imagine more complex interaction emerging when coupled interactions change the game state, creating the necessary context for future context-dependent interactions. For example, the pinball-like interaction in part B of scenario 2 could be expanded into a sequence where the player needs to time a second *Rotating Barrier* to hit the sliding *Push-Block* while it's moving. Or, a *Push-Block* could be positioned between two different colour *Rotating Barriers* and *Pressure Plates* such that it oscillates between the two, creating an on-going effect in the rest of the level.



B

1. **The *Push-Block* with no friction slides until it hits the red *Rotating Barrier*.**
2. **The *Push-Block* activates the red *Pressure Plate*, opening the red *Rotating Barrier*.**

Scenario #2: Part B

All in all, there are many possible interesting puzzles and challenges that can arise from this small selection of objects with limited properties, due to their deep relationships.